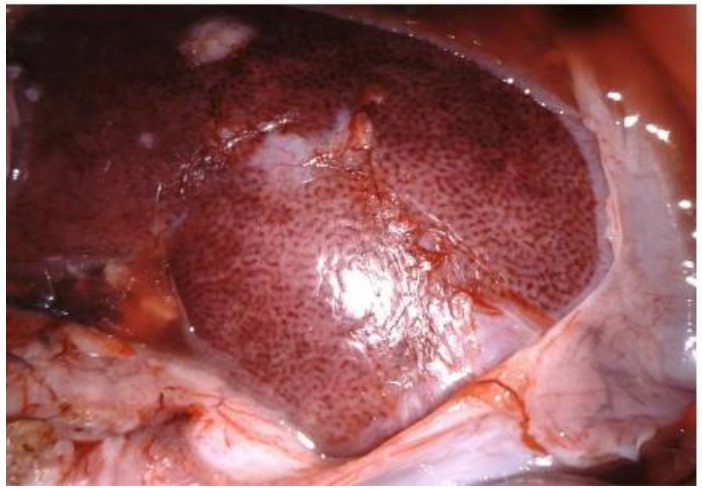
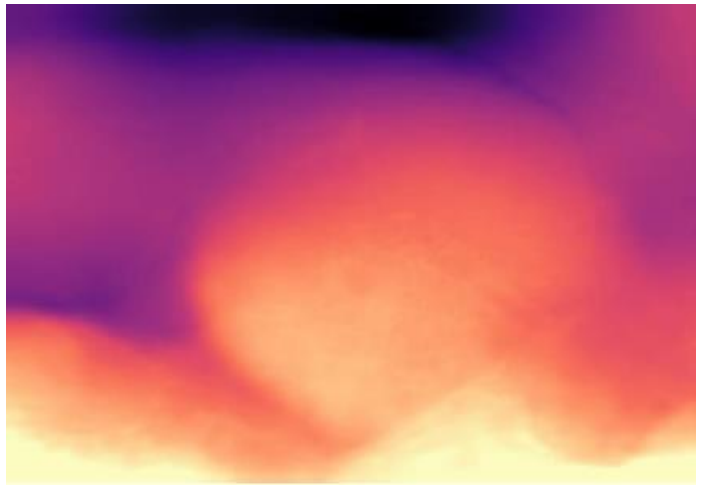
Single-Image Depth Estimation

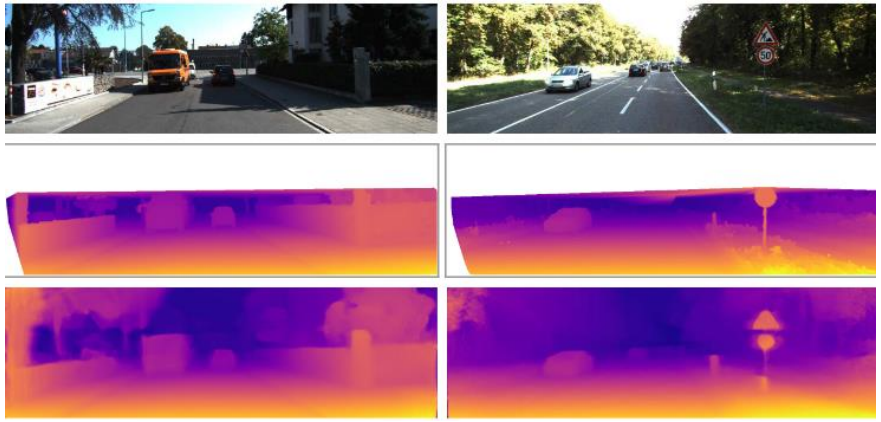
Literature Review

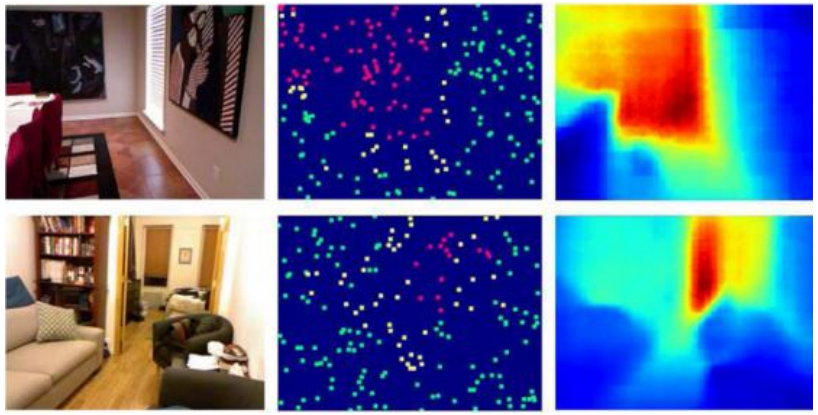




Data-driven approaches

- **Architectures→** CNN-based architectures
  - Supervised Approaches: IGNORED
  - Self-supervised approached: The majority
- **Loss function** → Established
- **Training datasets** → Dashcam videos, housing scenery





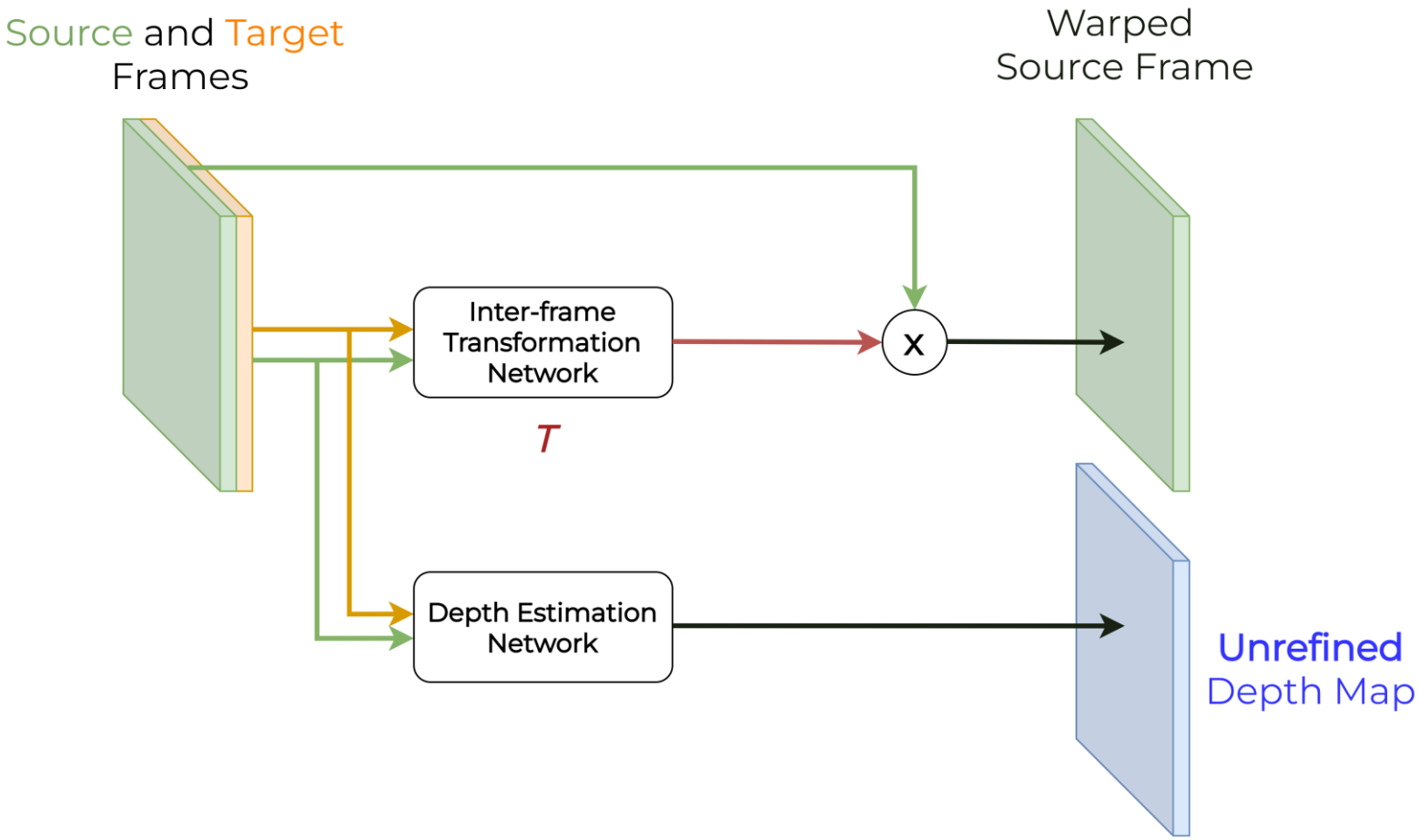
Data-driven approaches

- **SIDE for surgical images**
  - No public annotated datasets available
  - Established that the problem is **more complex** than outdoor scenery
    →Lack of geometry/sharp edges
    →Common non-Lambertian reflection phenomena
    →Limited camera motion
  - Self-Supervised approaches often rely on **temporal disparity** OR **spatial disparity** (stereo-pairs)
    → Supervisory signal from disparity maps "feature flow"

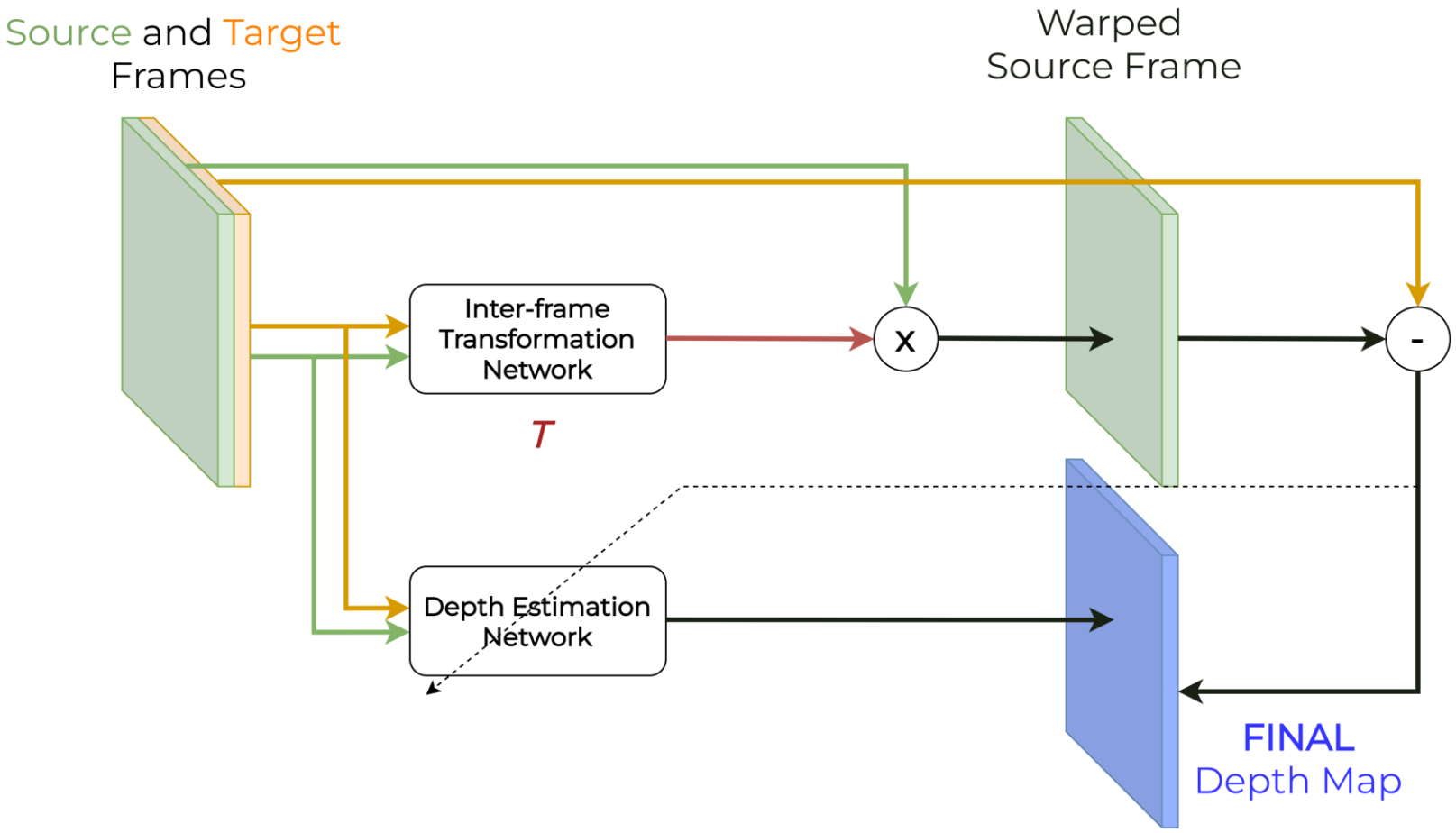
Data-driven approaches

- **State Of The Art Pipeline**
  1. From multi-frame information (usually *source* and *target* frame), estimate the camera motion transformation between frames and an **unrefined depth map**
  2. Use the predicted camera motion to warp the *source* frame to and obtain an estimated *target* frame
  3. Compare the predicted *target* frame with the real one and **refine the depth map**

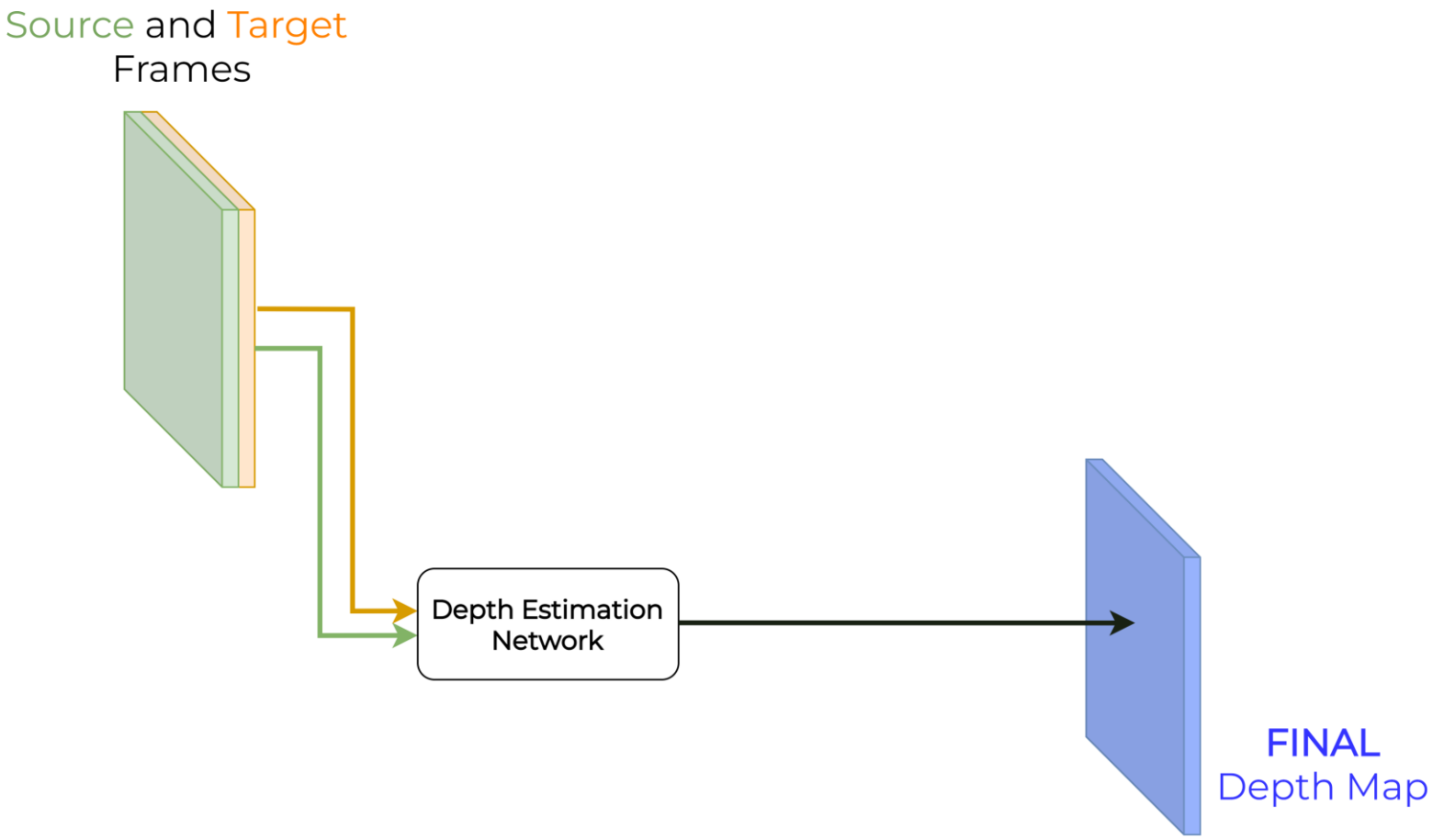
State Of The Art Pipeline: Training



State Of The Art Pipeline: Training



State Of The Art Pipeline: Inference



Non-data-driven approaches

- SLAM (V-SLAM)
- SfM (Structure From Motion)

Based on **feature extraction** → SPARSE depth map. Depth is estimated only on the feature points

→Heavily dependent on large camera ranges of motion

→Feature points may not be c

Considerations

- Loose inspiration from work on non-surgical images with adaptations

- Self-Supervision approach: **Stereo pair IS AVAILABLE** for training
  →Spatial & Temporal inter-frame correlation at training time
  →Temporal inter-frame correlation at inference time

- How to deal with the limited camera motion range? (if it is going to be a problem)